

Intel 80186 Microprocessor

1 Introduction

The **Intel 80186** is a 16-bit microprocessor belonging to the 80x86 family. It is an enhanced and upward-compatible version of the Intel 8086. The processor integrates several peripheral functions on-chip and is mainly used as an embedded controller.

2 Versions of Intel 80186

- **80C186XL**: Extended temperature range and enhanced speed
- **80C186EA**: Enhanced architecture
- **80C186EB**: Further refined version
- **80C186EC**: Enhanced and complete version

3 Key Specifications

- Data Bus Width: 16-bit
- Address Bus Width: 20-bit
- Maximum Addressable Memory: 1 MB
- Internal Register Structure: Same as Intel 8086
- Prefetch Queue Size: 6 bytes

4 Internal Architecture Overview

The Intel 80186 integrates several peripheral devices directly on the processor, greatly reducing the number of external support components required.

4.1 Major Functional Units

- Execution Unit (EU)
- Bus Interface Unit (BIU)
- On-chip Clock Generator
- Programmable Interrupt Controller
- Programmable Timers
- Programmable DMA Controller
- Programmable Chip Selection Unit
- Refresh Control Unit

5 Instruction Prefetch Queue

- Prefetch queue length: 6 bytes
- Enables overlap of instruction fetch and execution
- Improves overall instruction throughput

6 On-Chip Clock Generator

- Replaces the external 8284A clock generator
- Crystal frequency is twice the processor operating frequency
- CLKOUT pin provides system clock:
 - Frequency = half of crystal frequency
 - Duty cycle = 50%

7 Interrupt Handling Mechanism

7.1 Programmable Interrupt Controller (PIC)

- Internal PIC operates as the master controller
- Can control up to two external 8259A PICs as slaves
- Without external PICs, supports five interrupt inputs:
 - INT0 to INT3
 - Non-maskable interrupt (NMI)
- Additional interrupt vectors are provided for internal peripherals

8 Programmable Timer Unit

- Contains three fully programmable 16-bit timers

8.1 Timer Functions

- **Timer 0 and Timer 1**
 - Generate external waveforms
 - Driven by internal or external clock
 - Used to count external events
- **Timer 2**
 - Internal timer
 - Clocked by the master clock

- Generates interrupt after a specified number of clocks
- Can provide clock input to other timers

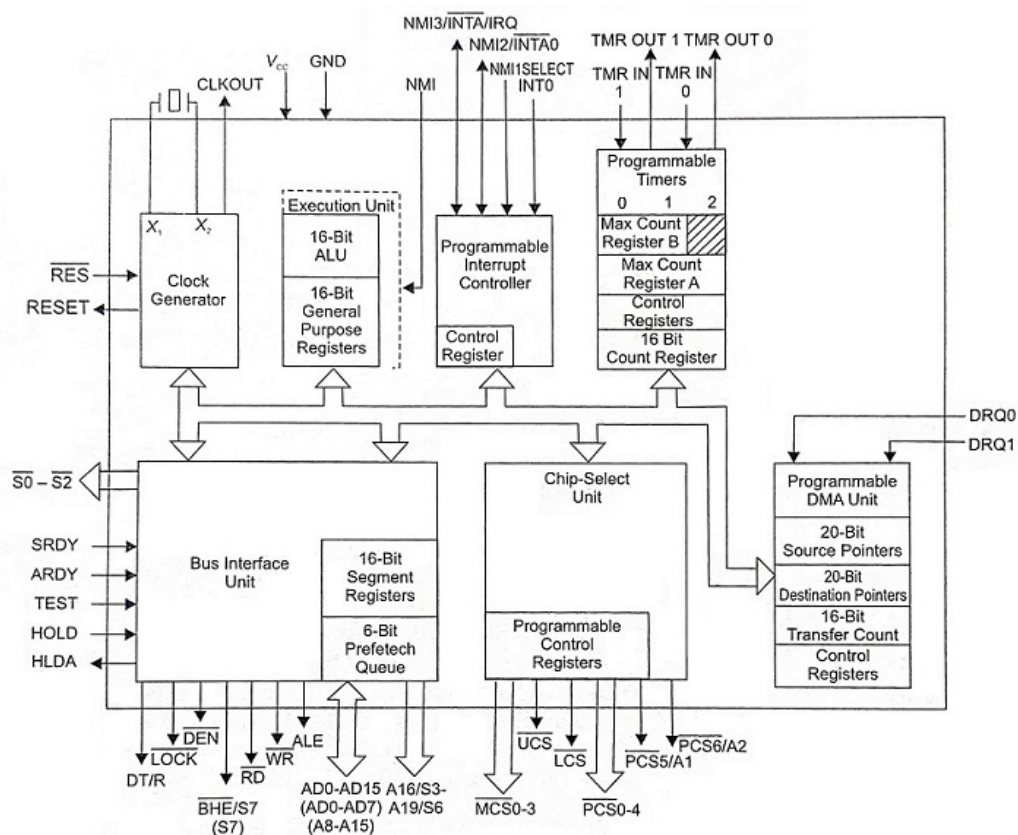


Fig. 11.1 Block diagram of Intel 80186/80188 architecture

Figure 1: Block Diagram of Intel 80186 Microprocessor

9 Programmable DMA Controller

- Contains two DMA channels
- Supports data transfer between:
 - Memory to memory
 - Memory and I/O
 - I/O devices

10 Programmable Chip Selection Unit

- Built-in programmable memory and I/O address decoder
- Provides:
 - Six output lines for memory selection
 - Seven output lines for I/O selection

11 Power Management Features

11.1 Power Save Mode

- Divides system clock by 4, 8, or 16
- Reduces power consumption
- Entered by software and exited by interrupt

11.2 Power Down Mode

- Stops the clock completely
- Entered by execution of the HLT instruction
- Exited by any interrupt

12 Refresh Control Unit

- Automatically refreshes dynamic RAM (DRAM)
- Generates refresh row address and control signal

13 Peripheral Control Block (PCB)

- Consists of 256 control registers
- Registers are 16-bit wide
- Controls all internal peripherals

13.1 PCB Addressing

- Default I/O address range: FF00H – FFFFH
- PCB can be relocated to memory or I/O space
- Relocation controlled using registers at offsets FEH and FFH

14 Interrupt System in Intel 80186

- Similar interrupt structure to Intel 8086
- Additional interrupt vectors for internal devices
- Supports special interrupts:
 - BOUND
 - Unused opcode
 - ESC opcode

15 Advantages of Intel 80186

- Greatly reduces the number of peripheral components
- Increases overall system utility
- Well suited for embedded controller applications
- Used in caching disk controllers, LAN controllers, and cellular network switching systems

16 Applications

- Embedded controllers
- Disk controllers
- LAN controllers
- Cellular telephone network switching systems

17 Difference Between Intel 8086 and Intel 80186

Feature	Intel 8086	Intel 80186
Processor Type	General-purpose microprocessor	Embedded controller-oriented processor
Clock Generator	External 8284A required	On-chip clock generator
Interrupt Controller	External 8259A required	Built-in PIC
Timers	Not available	Three 16-bit programmable timers
DMA Controller	External controller required	Built-in two-channel DMA
Chip Selection	External decoding required	Built-in programmable selection unit
Power Management	Not supported	Power-save and power-down modes
External Hardware	More support chips required	Fewer external components required
Applications	General computing systems	Embedded and control systems